

X-AXIS FIRST TEST - PHASE-BY-PHASE COMMISSIONING PROCEDURE

KOTEC KTPG-320 | UC300 (EC500) + Mach3 | X axis = YPV drive n1 | SOMIK Skikda

Machine:	KOTEC KTPG-320 CNC Plasma Cutting Machine
Axis under test:	X (gantry long travel) - left / right jog
Drive:	YPV n1 - YBL13S-75L.Z (1000 W, 5.5 A, 2000 rpm, encoder 8P / 2500)
Controller:	Digital Dream UC300 (EC500) - USB + Mach3 + DigitalDream plugin
Ballscrew pitch (X):	10 mm / rev
Steps per mm (X):	1000 (ELGN = ELGD = 10000)
Jog velocity:	2000 mm/min
Jog acceleration:	300 mm/s ²
Date:	
Technician:	
Result:	Pass / Fail

SAFETY: the plasma torch must remain OFF and disconnected from gas and power throughout all phases. Keep an extinguisher nearby and stay clear of the moving gantry. Cut mains + both 24V supplies before any wiring.

PHASE A - PREPARE (POWER OFF)

- Cut the electrical cabinet mains (220/380 V AC) and BOTH 24V DC supplies (System + IO).
- Verify with a multimeter (AC volts): 0 V between R-S, R-T, S-T on YPV drive n1.
- Confirm the charge LED on drive n1 is fully OFF before touching any terminal.
- Never touch terminals R, S, T, U, V, W while the charge LED is lit.
- Identify hardware: X drive = YPV n1, feeding the X motor YBL13S-75L.Z (1000 W, 5.5 A).
- Gather material: shielded twisted-pair cable (STEP/DIR, max 3 m), 2x 24V relays (coil < 100 mA), NC E-stop button, 24V PSU(s) ≥ 1 A (System + IO), screwdrivers, multimeter, wire strippers.

PHASE A - CHECKLIST

- ☐ Mains + both 24V supplies OFF
- ☐ R/S/T on drive n1 reads 0 V (multimeter)
- ☐ Charge LED on drive n1 fully OFF
- ☐ X drive = YPV n1 and X motor identified
- ☐ Material available (cable, relays, E-stop, PSUs, tools)

PHASE B - WIRE X ONLY (POWER STILL OFF)

B.1 - Power wiring

- Mains -> YPV drive n1 power terminals.
- 24V DC System (GND, 24V) -> UC300 System terminal block (≥ 1 A).
- 24V DC IO (COM-, COM+) -> UC300 IO terminal block (≥ 1 A).
- Drive logic power: 24G on CN1-5 (I/O common 24V).

B.2 - STEP / DIR from UC300 to drive (CN1)

- Wire COM- (UC300) -> DGND (CN1-31) FIRST - reference for all other signals (mandatory).

- XP+ -> CN1-26 (+PPCMD) / XP- -> CN1-27 (-PPCMD) [STEP]
- XD+ -> CN1-28 (+NPCMD) / XD- -> CN1-29 (-NPCMD) [DIR]
- Use shielded twisted-pair cable; separate pair for STEP and DIR.
- Ground the cable shield ONLY at the UC300 cabinet side (never both ends).

B.3 - SON / RST relay circuits

Never connect a UC300 output directly to CN1-6 or CN1-7 - a 24V relay is mandatory. Relay coil current must be < 100 mA (ULN2803A internal driver limit).

- SON relay - coil circuit (UC300 side): +24V DC -> coil pin 1 ... coil pin 2 -> OT1 (Spindle terminal, pin 3).
- SON relay - contact circuit (drive side, use NO contact): CN1-6 (/SON) -> COM ... NO -> CN1-5 (24G).
- RST relay - coil circuit (UC300 side): +24V DC -> coil pin 1 ... coil pin 2 -> OUT4 (DB9 General Output, pin 2).
- RST relay - contact circuit (drive side, use NO contact): CN1-7 (/RST) -> COM ... NO -> CN1-5 (24G).

B.4 - Motor + safety

- Motor power U / V / W between drive n1 and X motor (cable rated $\geq$ 5.5 A, recommended 8-10 A).
- E-Stop button wired NC (normally closed) -> IN01. Never wire it NO.

B.5 - Continuity check (power off)

- With multimeter continuity: verify CN1-5 (24G), CN1-6 (/SON), CN1-7 (/RST), CN1-26/27 (+PPCMD/-PPCMD), CN1-28/29 (+NPCMD/-NPCMD), CN1-31 (DGND).
- Confirm no continuity between COM- (UC300) and any motor power terminal (U/V/W).

PHASE B - CHECKLIST

- ☐ Mains -> drive n1; 24V System + IO -> UC300; 24G -> CN1-5
- ☐ COM- / DGND -> CN1-31 wired first (mandatory)
- ☐ XP+/XP- -> CN1-26/27 (STEP); XD+/XD- -> CN1-28/29 (DIR)
- ☐ Shield grounded at UC300 cabinet side only
- ☐ SON relay: OT1 -> coil; CN1-6 / NO / CN1-5
- ☐ RST relay: OUT4 -> coil; CN1-7 / NO / CN1-5
- ☐ Coil current of both relays < 100 mA
- ☐ Motor U/V/W connected; E-Stop NC -> IN01
- ☐ Continuity verified on CN1 pins; no short to U/V/W

PHASE C - YPV SETUP (X DRIVE ONLY, CN3 / RS-232)

Use JS YPV Setup software connected to CN3 (RS-232) of drive n1. These parameters are independent of the controller.

Reg.	Parameter	Value	Description	Priority
#1	CNTL	2	Control mode = Position (P mode)	CRITICAL
#2	PMOD	0	Pulse format = PULSE/DIR	CRITICAL
#32	ELGN	10000	Electronic gear numerator	CRITICAL
#36	ELGD	10000	Electronic gear denominator (steps/mm = 1000)	CRITICAL
#44	RPM	2000	Motor rated speed (nameplate)	IMPORTANT
#45	ENCO	2500	Encoder resolution (points/rev)	IMPORTANT
#43	MTYP	8	Motor pole count (8P)	IMPORTANT
#7	PACC	0	Accel/decel time (Mach3 handles the ramp)	IMPORTANT

#53	PERZ	5000	Following-error limit	USEFUL
#59	MRO	1	Rotation direction (1=CW, 0=CCW) - verify on test	USEFUL

- Open JS YPV Setup and connect to drive n1 via CN3.
- Enter the values from the table above.
- CLICK 'Upload and Program' to write to FLASH ROM - otherwise settings are lost on power off.

PHASE C - CHECKLIST

- ☐ CNTL = 2 (Position mode)
- ☐ PMOD = 0 (PULSE/DIR)
- ☐ ELGN = 10000 / ELGD = 10000
- ☐ RPM = 2000 / ENCO = 2500 / MTYP = 8 / PACC = 0
- ☐ PERZ = 5000 / MRO = 1
- ☐ 'Upload and Program' executed (FLASH ROM)

PHASE D - MACH3 + PLUGIN SETUP (PC)

D.1 - Install plugin and create profile

- Copy DigitalDream.dll into C:\Mach3\PlugIns.
- Launch Mach3. In 'Session Profile' click 'Create Profile', name it 'DigitalDream', check 'Default Profile Values', OK.
- In the hardware selection window choose 'DigitalDream-DigitalDream-PlugIn' (NOT 'Normal Printer port').
- Verify the message 'UC300 device is connected to your computer'.
- Config > Config Plugins: confirm DigitalDream-PlugIn is checked (enabled).

D.2 - Motor Outputs (Config > Ports & Pins > Motor Outputs)

- X axis: check 'Enabled'. Step Pin# = 2, Dir Pin# = 3 (Port No. 2).
- Click Apply. Do NOT check Dir LowActive yet (adjust after direction test).
- Click 'Save Axis Settings'.

D.3 - Output Signals (Config > Ports & Pins > Output Signals)

- Map OT1 / OUT1 (Spindle port, pin 3) to the output used for X SON - drive it via Mach3 spindle enable (M3) or an Output Signal on Port 2, pin 1.
- Map OUT4 (DB9 pin 2) to RST X - Port 2, pin 4.

UC300 outputs are defined on Port No. 2 by default. Keep wiring and Mach3 mapping in sync.

D.4 - Input Signals (Config > Ports & Pins > Input Signals)

- Map E-Stop -> IN01 (Port 2, pin 1). Check 'Emulated' if required by the plugin version.

D.5 - Motor Tuning (Config > Motor Tuning)

- Select X axis. Steps per Unit = 1000. Velocity = 2000 mm/min. Acceleration = 300 mm/s².
- Click 'SAVE AXIS SETTINGS' before moving on.

PHASE D - CHECKLIST

- ☐ DigitalDream.dll installed in C:\Mach3\PlugIns
- ☐ Profile 'DigitalDream' created with DigitalDream-PlugIn
- ☐ UC300 'device connected' message shown
- ☐ X Motor Output: Step 2 / Dir 3, Port 2, Enabled
- ☐ SON X mapped to OT1/OUT1 (port 2 pin 1)
- ☐ RST X mapped to OUT4 (port 2 pin 4)
- ☐ E-Stop mapped to IN01
- ☐ Motor Tuning X: 1000 steps/unit, 2000 mm/min, 300 mm/s² - saved

PHASE E - POWER UP AND FIRST JOG

Torch still OFF gas + power. Move slowly. Keep hands clear of the gantry.

- Power up 24V System, then 24V IO, then mains. Confirm the charge LED lights on drive n1.
- In Mach3 confirm the UC300 connection is active ('device is connected').
- TEST E-STOP: press the E-stop button - an alarm must appear immediately in Mach3. Reset it.
- Enable X SON (turn on spindle / SON X output): drive n1 must switch to 'Servo ON'.
- If the drive does not enable: check relay coil circuit, DGND / COM- on CN1-31, continuity CN1-6 <-> CN1-5, and coil current < 100 mA.
- Jog X a few mm at low feed: jog LEFT then jog RIGHT (Tab jog or arrow keys).
- Check the direction matches the command.
- If reversed: correct ONLY ONE setting - either MRO in YPV Setup or Dir LowActive in Mach3 - never both at the same time. Save Axis Settings after any change.
- Verify both left and right movement at low speed, then gradually increase to the target jog speed.

PHASE E - CHECKLIST

- ☐ Power up sequence OK (24V System -> IO -> mains)
- ☐ Charge LED on drive n1 lit
- ☐ UC300 connected in Mach3
- ☐ E-Stop tested (alarm appears / machine stops)
- ☐ X drive in 'Servo ON' when SON enabled
- ☐ X jog LEFT works
- ☐ X jog RIGHT works
- ☐ Direction correct (or corrected with only ONE setting)
- ☐ No alarm / following-error during jog

PHASE F - CALIBRATION SANITY (OPTIONAL)

- In Mach3 jog X a known distance, e.g. 100 mm.
- Measure the actual travel with a tape measure.
- If error: new ELGD = old ELGD x (measured distance / commanded distance).
- Reprogram ELGD in YPV Setup AND update Steps per Unit in Mach3 Motor Tuning (both must stay consistent). 'Upload and Program' the drive.
- Repeat until the error is < 0.1 mm on 100 mm.

PHASE F - CHECKLIST

- ☐ 100 mm jog performed

- ☐ Measured travel recorded: _____ mm
- ☐ ELGD / Steps per Unit corrected if needed
- ☐ Error < 0.1 mm on 100 mm

RESULT

Result	Signature
Pass / Fail	